

SAT Unit 2.3.1

Polynomial and Rational Functions

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Polynomial Function

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Definition of Polynomial Functions

Definition of Polynomial:

An expression of the form:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where:

- n is a nonnegative integer (called the **degree**).
- $a_n, a_{n-1}, \dots, a_0$ are real numbers (called **coefficients**).

Terminology:

- **Monomial:** A polynomial with one term (e.g., $5x^3$)
- **Binomial:** Two unlike terms (e.g., $x^2 + 3x$)
- **Trinomial:** Three unlike terms (e.g., $x^2 + 3x + 2$)
- **Like Terms:** Terms that have the same variable and exponent (e.g., $3x^2$ and $-2x^2$)

Structure and Degree of Polynomials

Degree:

- The **degree of a monomial** is the sum of the exponents of all variables in that term.
- The **degree of a polynomial** is the greatest degree among all its terms.

Components of a Polynomial:

- **Leading Term:** The term with the highest degree.
- **Leading Coefficient:** The coefficient of the leading term.
- **Constant Term:** A term with no variable (degree 0).

Example:

$$f(x) = 3x^3 - 4x^2 + 7x + 18$$

- Degree: 3 Leading Coefficient: 3 Constant: 18

Zeros of Polynomial Functions and Graph Behavior

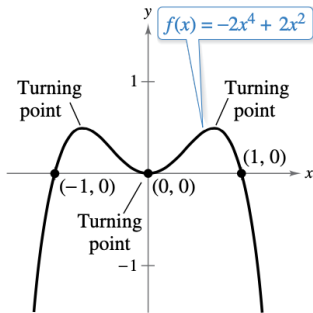
Real Zeros of Polynomial Functions

When f is a polynomial function and a is a real number, the following are equivalent:

- 1 $x = a$ is a **zero** of f .
- 2 $x = a$ is a **solution** of $f(x) = 0$.
- 3 $(x - a)$ is a **factor** of $f(x)$.
- 4 $(a, 0)$ is an **x-intercept** of the graph of f .

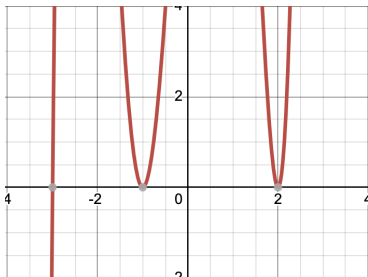
Repeated Zeros and Multiplicity

- A factor $(x - a)^k$, $k > 1$, yields a **repeated zero** $x = a$ of **multiplicity** k :
 - If k is odd: the graph **crosses** the x-axis.
 - If k is even: the graph **touches** the x-axis.



Question: Polynomial Matches the Graph

The graph of a function f is shown. Which of the following functions could define f ?



Choices:

- A. $f(x) = (x - 3)(x - 1)^2(x + 2)^2$
- B. $f(x) = (x - 3)^2(x - 1)(x + 2)$
- C. $f(x) = (x + 3)(x + 1)^2(x - 2)^2$
- D. $f(x) = (x + 3)^2(x + 1)(x - 2)$

Answer: Identifying a Polynomial from Its Graph

Goal: Match the graph of $f(x)$ to one of the given polynomial equations.

Step 1: Identify the Zeros

- The graph crosses/touches the x -axis at:

$$x = -3, \quad x = -1, \quad x = 2$$

Step 2: Analyze Multiplicities

- Touches at $x = -1$ and $x = 2 \Rightarrow$ even multiplicity (likely squared).
- Crosses at $x = -3 \Rightarrow$ odd multiplicity (likely linear).

Step 3: Determine Equation

$$f(x) = (x + 3)(x + 1)^2(x - 2)^2$$

Correct Answer: C

Question: Solving a Factored Equation with Radicals

$$\frac{5}{8}(5x + 8) \left(x + \sqrt{5k + 8} \right) \left(x - \sqrt{5k + 8} \right) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 64. What is the value of k ?

Answer and Explanation

We are given:

$$\frac{5}{8}(5x + 8)(x + \sqrt{5k + 8})(x - \sqrt{5k + 8}) = 0$$

The three roots come from:

- $5x + 8 = 0 \Rightarrow x = -\frac{8}{5}$
- $x + \sqrt{5k + 8} = 0 \Rightarrow x = -\sqrt{5k + 8}$
- $x - \sqrt{5k + 8} = 0 \Rightarrow x = \sqrt{5k + 8}$

So the roots are:

$$x = -\frac{8}{5}, \quad -\sqrt{5k + 8}, \quad \sqrt{5k + 8}$$

Let's compute the product of the roots (since the coefficient $\frac{5}{8}$ does not affect the roots):

$$\left(-\frac{8}{5}\right) \cdot \left(-\sqrt{5k + 8}\right) \cdot \left(\sqrt{5k + 8}\right) = \frac{8}{5}(5k + 8)$$

Answer and Explanation

We're told the product of the solutions is 64, so:

$$\frac{8}{5}(5k + 8) = 64 \Rightarrow (5k + 8) = \frac{64 \cdot 5}{8} = 40 \Rightarrow 5k = 32 \Rightarrow k = \boxed{6.4}$$

Correct Answer: $\boxed{6.4}$

Polynomial Long Division

Purpose: Polynomial long division is used to divide a polynomial by another polynomial of lower or equal degree.

Steps:

- 1 Divide the leading term of the dividend by the leading term of the divisor.
- 2 Multiply the divisor by the result and subtract from the dividend.
- 3 Repeat the process with the remainder.

Polynomial Long Division

$$\begin{array}{r} 2x^2 + 7x + 9 \\ x - 2 \overline{) 2x^3 + 3x^2 - 5x + 6} \\ \underline{-2x^3 + 4x^2} \\ 7x^2 - 5x \\ \underline{-7x^2 + 14x} \\ 9x + 6 \\ \underline{-9x + 18} \\ 24 \end{array}$$

Steps:

- 1 Divide
- 2 Multiply
- 3 Subtract

Result:

$$\frac{2x^3 + 3x^2 - 5x + 6}{x - 2} = 2x^2 + 7x + 9 + \frac{24}{x - 2}$$

Remainder Theorem

Theorem Statement: If a polynomial $f(x)$ is divided by $(x - c)$, then the remainder is:

$$\text{Remainder} = f(c)$$

Why this works: According to the Division Algorithm:

$$f(x) = (x - c)q(x) + r$$

where:

- $q(x)$ is the quotient
- r is the remainder

Substitute $x = c$:

$$f(c) = (c - c)q(c) + r = 0 + r = r \Rightarrow \boxed{f(c) = \text{Remainder}}$$

Key Use: Quickly evaluate whether $(x - c)$ is a factor of $f(x)$ — if $f(c) = 0$, then $(x - c)$ is a factor!

Question: Polynomial Divisibility

The polynomials are defined as:

$$f(x) = 2x^3 + 6x^2 + 4x, \quad g(x) = x^2 + 3x + 2$$

Which of the following polynomials is divisible by $2x + 3$?

- A. $h(x) = f(x) + g(x)$
- B. $p(x) = f(x) + 3g(x)$
- C. $r(x) = 2f(x) + 3g(x)$
- D. $s(x) = 3f(x) + 2g(x)$

Answer and Explanation

To test divisibility by $2x + 3$, substitute $x = -\frac{3}{2}$. If the expression evaluates to 0, the polynomial is divisible by $2x + 3$.

Try option B:

$$p(x) = f(x) + 3g(x) \Rightarrow p\left(-\frac{3}{2}\right) = \frac{3}{4} + 3 \cdot \left(-\frac{1}{4}\right) = \frac{3}{4} - \frac{3}{4} = 0$$

Correct Answer: B

Question: Remainder Theorem and Polynomial Evaluation

For a polynomial $p(x)$, the value of $p(3) = -2$. Which of the following must be true about $p(x)$?

- A. $x - 5$ is a factor of $p(x)$.
- B. $x - 2$ is a factor of $p(x)$.
- C. $x + 2$ is a factor of $p(x)$.
- D. The remainder when $p(x)$ is divided by $x - 3$ is -2 .

Answer and Explanation

We are told that $p(3) = -2$. According to the Remainder Theorem, the remainder when a polynomial $p(x)$ is divided by $x - a$ is $p(a)$.

- Since $p(3) = -2$, the remainder when dividing $p(x)$ by $x - 3$ is -2 .
- This directly matches option **D**.
- Options A, B, and C all refer to whether a linear binomial is a factor of $p(x)$, which would require the remainder to be 0 at that root, but we are only given a non-zero value at $x = 3$.

Correct Answer: D



Rational Function

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Introduction to Rational Functions

Definition:

A **rational function** is a quotient of two polynomial functions, written in the form:

$$f(x) = \frac{N(x)}{D(x)}$$

where:

- $N(x)$ and $D(x)$ are polynomials,
- $D(x) \neq 0$.

Domain: The domain includes all real numbers **except** those for which $D(x) = 0$ (division by zero is undefined).

Question: Behavior of Rational Function

The function is defined as:

$$p(x) = \frac{2x - 1}{(x - 4)^2 - 6(x - 4) + 9}$$

For what value of x can the function $p(x)$ **not** cross the x -axis?

Answer and Explanation

To determine when $p(x)$ cannot cross the x -axis, we need to find when the numerator $2x - 1 = 0$ has no solution due to the denominator restricting the domain.

But since the function crosses the x -axis when $p(x) = 0$, this only happens when the numerator is zero and the denominator is non-zero.

So, we look for values of x where the denominator is undefined or causes a vertical asymptote, because these are values where the function can never cross the x -axis.

Denominator:

$$(x - 4)^2 - 6(x - 4) + 9$$

Let $u = x - 4$, then:

$$u^2 - 6u + 9 = (u - 3)^2 \Rightarrow (x - 7)^2$$

So denominator becomes zero when:

$$x - 7 = 0 \Rightarrow x = 7$$

Final Answer: 7

Undefined Values of Rational Functions

Key Concept: A rational function is undefined at values of x that make the denominator equal to zero.

General Form:

$$f(x) = \frac{N(x)}{D(x)}$$

- $N(x)$ is the numerator
- $D(x)$ is the denominator
- The function is undefined where $D(x) = 0$

Example:

$$f(x) = \frac{4}{x - 5}$$

- Set denominator equal to zero: $x - 5 = 0$
- Solution: $x = 5$
- Therefore, $f(x)$ is undefined at $x = 5$

Practices

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Question: Intercepts of a Polynomial Function

The graph of $y = f(x)$ has a y -intercept at $(0, 12)$ and x -intercepts at $(-3, 0)$ and $(2, 0)$. Which of the following equations could define f ?

A $f(x) = (x + 3)^2(x - 2)$

B $f(x) = (x + 3)(x - 2)^2$

C $f(x) = (x - 3)^2(x + 2)$

D $f(x) = (x - 3)(x + 2)^2$

Answer: Intercepts of a Polynomial Function

We are given:

- x -intercepts at $x = -3$ and $x = 2$, so the factored form must contain $(x + 3)$ and $(x - 2)$.
- y -intercept is at $(0, 12)$.

Try Option A:

$$f(x) = (x + 3)^2(x - 2)$$

At $x = 0$:

$$f(0) = (0 + 3)^2(0 - 2) = 9 \cdot (-2) = -18 \neq 12$$

Try Option B:

$$f(x) = (x + 3)(x - 2)^2$$

At $x = 0$:

$$f(0) = (0 + 3)(0 - 2)^2 = 3 \cdot 4 = 12$$

Correct!

Question: Distinct Zeros of a Polynomial

$$f(x) = -x(x - a)(x + a)(x + b)(x + b)$$

In the given function f , a and b are positive constants and $a \neq b$. For how many distinct values of x does $f(x) = 0$?

- A Two
- B Three
- C Four
- D Five

Answer: Distinct Zeros of a Polynomial

We are solving for the number of **distinct** values of x such that:

$$f(x) = -x(x - a)(x + a)(x + b)(x + b)$$

Set each factor equal to zero:

$$x = 0, x = a, x = -a, x = -b$$

(Note: $x + b$ appears twice, but it yields the same solution $x = -b$.)
Since $a \neq b$ and both are positive, all four roots are distinct.

Correct answer: C) Four

Question: Behavior of Rational Function

The function is defined as:

$$p(x) = \frac{3x + 2}{(x - 1)^2 - 4(x - 1) + 4}$$

For what value of x can the function $p(x)$ **not** cross the x -axis?

Answer: Behavior of Rational Function

We are given:

$$p(x) = \frac{3x + 2}{(x - 1)^2 - 4(x - 1) + 4}$$

The graph cannot cross the x -axis where the denominator is zero. Simplify the denominator:

$$(x - 1)^2 - 4(x - 1) + 4$$

Let $u = x - 1$, then:

$$u^2 - 4u + 4 = (u - 2)^2 = (x - 3)^2$$

So the denominator is zero when $x = 3$.

Thus, $p(x)$ **cannot** cross the x -axis at:

$$\boxed{x = 3}$$

Question: Polynomial Evaluation and Remainder Theorem

For a polynomial $q(x)$, the value of $q(4) = 7$. Which of the following must be true about $q(x)$?

- A $x - 4$ is a factor of $q(x)$.
- B $x + 4$ is a factor of $q(x)$.
- C The remainder when $q(x)$ is divided by $x - 4$ is 7.
- D The remainder when $q(x)$ is divided by $x + 4$ is 7.

Answer: Polynomial Evaluation and Remainder Theorem

Given that $q(4) = 7$, by the **Remainder Theorem**, we know that:

The remainder when $q(x)$ is divided by $x - 4$ is $q(4) = 7$.

So the correct statement is:

C) The remainder when $q(x)$ is divided by $x - 4$ is 7.

Correct answer: C

Thank You!

We appreciate your time and focus.

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